SUNIL BELDE

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Dear Hiring Manager,

I am a machine learning engineer at Capital One, where I build GenAI systems that do real work inside the business: agents that reason over enterprise data, and pipelines that turn plain-language instructions into production code. I hold an M.S. in Computer Science from the University of Illinois Chicago, and I care most about systems that can be measured, trusted and run every day.

My main project is a human-in-the-loop mapping engine for the Discover File System migration. Built on the ReAct pattern, it proposes source-to-target mappings, critiques them and refines them, drawing on requirements documents and past conversion knowledge. Through enterprise MCP tools such as Glean and Ask-Data it looks up registered data assets before it decides. I treat evaluation as part of the product: I ran benchmark evaluations comparing models, reasoning settings and prompt versions, fed the results back into the agent's prompts and configuration, and the assistant reached 75% source-field recall on cross-domain benchmarks.

I also architected a GenAI ETL platform that translates natural-language mapping instructions into PySpark code deployable as AWS Glue jobs. It pairs Llama 3.3-70B, LangChain and SimSearch retrieval with deterministic Python and NetworkX DAG orchestration, automated code validation and retry loops. It reduced engineering effort by over 70% and cut delivery time from about 5 months to 4 weeks. To pick its production model, I led the benchmarking of Mixtral, Llama, GPT-OSS 120B and other open models on latency, accuracy, hallucination rate and cost.

Before Capital One, I built document intelligence pipelines at Dockskiff that combined NER, OCR and PyTorch models to extract contract metadata, including an auto-labeling pipeline with Prodigy that cut manual annotation effort by 60%. Earlier still, at Acheron, I migrated legacy systems to Kubernetes with Helm and cut infrastructure costs by 15%.

I would welcome the chance to talk about how this experience could help your team. Thank you for your time and consideration.

Sincerely,
Sunil Belde